

Strip-growth obstructions for rank-one truncated Toeplitz symbols

A partial answer to Conjectures 3.1–3.2 of arXiv:2107.09115

Automated mathematical-discovery run `fa_banach_001`

13 August 2026

Status. Partial result, likely valid, subject to human review.

1 Source conjecture and scope

For an inner function I , put $K_I^2 = H^2 \ominus IH^2$. If I has an angular derivative at $\zeta \in \mathbb{T}$, then evaluation at ζ is bounded on K_I^2 , with reproducing kernel

$$k_\zeta^I(z) = \frac{1 - \overline{I(\zeta)}I(z)}{1 - \bar{\zeta}z}.$$

The rank-one map $k_\zeta^I \otimes k_\zeta^I$ is a bounded truncated Toeplitz operator.

On page 10, O’Loughlin asks the following equivalent conjectures [1]:

Every bounded truncated Toeplitz operator on K_I^2 has a bounded symbol if and only if I is one-component.

Every compact truncated Toeplitz operator on K_I^2 has a symbol in $IC(\mathbb{T})$ if and only if I is one-component.

A long standing open conjecture regarding symbols of bounded truncated Toeplitz operators is the following.

Conjecture 3.1. Every bounded truncated Toeplitz operator on K_I^2 has a bounded symbol if and only if I is one-component.

We can now state an equivalent formulation of the above conjecture, which is the following.

Conjecture 3.2. Every compact truncated Toeplitz operator on K_I^2 has a symbol in $IC(\mathbb{T})$ if and only if I is one-component.

The forward implication for one-component functions is known [4]. The converse remains open. The result below proves a new endpoint obstruction and applies it to two explicit non-one-component inner functions. It does not settle the universal converse.

2 Proof intuition

Sarason’s criterion for the boundary rank-one operator says that a bounded symbol would allow one to correct k_ζ^I by a function Ih , $h \in H^2$, so that the resulting analytic function has bounded real part. Such a function maps the disk into a vertical strip and can grow only logarithmically toward the boundary.

Now approach ζ very tangentially through points z_n at which $I(z_n)$ is tiny. The correction $I(z_n)h(z_n)$ is negligible by the H^2 point-evaluation estimate, whereas the kernel is essentially $(1 - \bar{\zeta}z_n)^{-1}$. If that reciprocal grows faster than the permitted logarithm, no bounded symbol can exist. Zeros make the correction vanish exactly; tiny values near atoms of a singular measure give a zero-free version of the same mechanism.

3 The strip obstruction

Lemma 1 (Logarithmic growth in a vertical strip). *If $F \in H^2$ and $\operatorname{Re} F \in L^\infty(\mathbb{T})$, then a constant $C_F < \infty$ exists such that*

$$|F(z)| \leq C_F \log \frac{e}{1 - |z|}, \quad z \in \mathbb{D}.$$

Proof. Let $M = \|\operatorname{Re} F\|_\infty$. The Poisson representation gives $|\operatorname{Re} F(z)| \leq M$ in $\mathbb{D}$. If $M = 0$, then F is a purely imaginary constant. Otherwise put

$$G = \frac{F - i \operatorname{Im} F(0)}{M}.$$

Then $|\operatorname{Re} G| \leq 1$ and $\operatorname{Im} G(0) = 0$. Carathéodory's inequality, applied to either $1 + G$ or $1 - G$, gives

$$|G'(z)| \leq \frac{4}{1 - |z|^2}.$$

Integrating on the radius from 0 to $z = re^{it}$ yields

$$|G(z) - G(0)| \leq 2 \log \frac{1 + r}{1 - r}.$$

Since $|G(0)| \leq 1$, the stated estimate follows after absorbing M and $|\operatorname{Im} F(0)|$ into C_F . $\square$

We use the following individual rank-one criterion of Sarason [2, Section 5]; see also [3, Remark 5.4]. If I has an angular derivative at ζ , then $k_\zeta^I \otimes k_\zeta^I$ has a bounded symbol only if there is $h \in H^2$ such that

$$\operatorname{Re}(k_\zeta^I + Ih) \in L^\infty(\mathbb{T}). \quad (1)$$

Theorem 2 (Tangential small-value obstruction). *Let I be inner and suppose that I has an angular derivative at $\zeta \in \mathbb{T}$. Assume that points $z_n \in \mathbb{D}$ satisfy, with $d_n = |1 - \bar{\zeta}z_n|$,*

$$z_n \rightarrow \zeta, \quad I(z_n) \rightarrow 0, \quad (2)$$

$$\frac{|I(z_n)|}{\sqrt{1 - |z_n|^2}} = o(d_n^{-1}), \quad (3)$$

$$d_n \log \frac{e}{1 - |z_n|} \rightarrow 0. \quad (4)$$

Then the bounded rank-one truncated Toeplitz operator $k_\zeta^I \otimes k_\zeta^I$ has no bounded symbol. Consequently it has no symbol in $IC(\mathbb{T})$.

Proof. Suppose a bounded symbol exists. By (1), choose $h \in H^2$ and set $F = k_\zeta^I + Ih$ with bounded real part. Lemma 1 gives

$$|F(z_n)| \leq C_F \log \frac{e}{1 - |z_n|}. \quad (5)$$

Since $I(z_n) \rightarrow 0$,

$$|k_\zeta^I(z_n)| = \frac{|1 - \overline{I(\zeta)}I(z_n)|}{d_n} \geq \frac{1}{2d_n}$$

for all large n . The Hardy-space point-evaluation bound and (3) give

$$|I(z_n)h(z_n)| \leq \frac{\|h\|_2 |I(z_n)|}{\sqrt{1-|z_n|^2}} = o(d_n^{-1}).$$

Thus $|F(z_n)| \geq (3d_n)^{-1}$ eventually. This contradicts (5) and (4). $\square$

4 A Blaschke example beyond every finite Hardy-power test

Theorem 3 (All finite Hardy powers, but no bounded symbol). *For $n \geq 2$, let*

$$\varepsilon_n = e^{-\sqrt{n}}, \quad a_n = (1 - \varepsilon_n)e^{i/n},$$

and let B be the Blaschke product with zeros (a_n) . Then B has an angular derivative at 1 and

$$k_1^B \in H^p \quad \text{for every finite } p > 1,$$

but $k_1^B \otimes k_1^B$ has no bounded symbol.

Proof. The series $\sum_n \varepsilon_n$ converges, so B is well defined. Since $\varepsilon_n = o(n^{-1})$,

$$|1 - a_n| \asymp n^{-1}, \quad 1 - |a_n|^2 \asymp \varepsilon_n.$$

Hence, for every fixed finite $p > 1$,

$$\sum_{n \geq 2} \frac{1 - |a_n|^2}{|1 - a_n|^p} \asymp \sum_{n \geq 2} e^{-\sqrt{n}} n^p < \infty. \quad (6)$$

The case $p = 2$ is the Ahern–Clark angular-derivative condition, while the general case and the boundary-kernel criterion give $k_1^B \in H^p$ [5, 6]; see also [3, Section 2.2].

Use $z_n = a_n$ in Theorem 2. The first two small-value conditions are immediate because $B(a_n) = 0$, and

$$|1 - a_n| \log \frac{e}{1 - |a_n|} \asymp \frac{1 + \sqrt{n}}{n} \longrightarrow 0.$$

The theorem proves the assertion. $\square$

The known fixed-power obstruction says that failure of $k_\zeta^I \in H^p$ for one $p > 2$ rules out a bounded symbol [3, Theorem 5.3]. Estimate (6) shows that this test is silent for every finite p in Theorem 3.

5 A zero-free atomic singular example

For a finite positive singular measure μ on $\mathbb{T}$, write

$$S_\mu(z) = \exp \left(- \int_{\mathbb{T}} \frac{\xi + z}{\xi - z} d\mu(\xi) \right).$$

Theorem 4 (Zero-free version). *Let*

$$\xi_n = e^{i/n}, \quad c_n = e^{-\sqrt{n}}, \quad \mu = \sum_{n \geq 2} c_n \delta_{\xi_n},$$

and put $S = S_\mu$. Then S is zero-free, has an angular derivative at 1, and $k_1^S \in H^p$ for every finite $p > 1$. Nevertheless, $k_1^S \otimes k_1^S$ has no bounded symbol.

Proof. The measure is finite. For every finite $p > 1$,

$$\int_{\mathbb{T}} \frac{d\mu(\xi)}{|1 - \xi|^p} = \sum_{n \geq 2} \frac{c_n}{|1 - \xi_n|^p} \asymp \sum_{n \geq 2} e^{-\sqrt{n}} n^p < \infty.$$

As above, the case $p = 2$ gives the angular derivative and the general Ahern–Clark/Cohn criterion gives $k_1^S \in H^p$.

Set $r_n = 1 - c_n^2$ and $z_n = r_n \xi_n$. Because $c_n^2 = o(n^{-1})$,

$$d_n := |1 - z_n| \asymp n^{-1}, \quad \log \frac{e}{1 - |z_n|} = 1 + 2\sqrt{n}.$$

Thus (4) holds. The atom at ξ_n gives

$$\begin{aligned} |S(z_n)| &= \exp \left(- \int_{\mathbb{T}} \frac{1 - |z_n|^2}{|\xi - z_n|^2} d\mu(\xi) \right) \\ &\leq \exp \left(-c_n \frac{1 + r_n}{1 - r_n} \right) \leq \exp(-c_n^{-1}). \end{aligned}$$

In particular $S(z_n) \rightarrow 0$, and

$$\frac{|S(z_n)|}{\sqrt{1 - |z_n|^2}} \lesssim \frac{e^{-1/c_n}}{c_n} = o(n) = o(d_n^{-1}).$$

Theorem 2 completes the proof. □

6 Limitations, upgrade attempts, and novelty bounds

Theorems 3 and 4 provide explicit compact rank-one witnesses with no $IC(\mathbb{T})$ symbol and bounded rank-one witnesses with no bounded symbol. They therefore verify the conjectured negative direction for two non-one-component functions. They do not show that every non-one-component inner function has the tangential sequence required by Theorem 2.

Five focused routes were pursued. First, the exact Carleson-class condition $\mathcal{C}_1(I^2) = \mathcal{C}_2(I^2)$ was compared with Aleksandrov’s characterization by equality of all embedding classes; no valid exponent bootstrap was found. Second, point-mass and Clark-measure tests were used to seek the boundary derivative criterion, but the basic estimates reduce to the Clark isometry. Third, Sarason’s rank-one criterion yielded Theorem 2. Fourth, the result was upgraded first to the Blaschke example and then to the zero-free singular example, both beyond every finite H^p test. Fifth, product and model-space direct-sum constructions were tested as a route to a counterexample to the full conjecture; the needed H^1 endpoint decomposition was not obtained.

A bounded search through 13 August 2026 used arXiv, exact-title and exact-conjecture searches, the source citations, the phrases “bounded symbol one-component”, “rank-one truncated Toeplitz”, “weak factorization model space”, and the later survey/lecture literature. Recent sources still state the universal converse as open. The 2010 paper [3] proves the fixed- p necessary condition, but no located source states the strip-growth criterion or either all-finite- H^p example above. Novelty is plausible but provisional pending expert review.

7 Verification notes

The only non-elementary inputs are Sarason’s individual rank-one criterion and the Ahern–Clark/Cohn boundary-kernel criterion. Lemma 1 is a direct Carathéodory estimate. In both examples the series estimates dominate every polynomial because $e^{-\sqrt{n}}$ does. The singular estimate uses only the contribution of one positive atom, so there is no cancellation. No numerical evidence is used.

References

- [1] R. O’Loughlin, *Symbols of compact truncated Toeplitz operators*, J. Math. Anal. Appl. 507 (2022), 125819; arXiv:2107.09115.
- [2] D. Sarason, *Algebraic properties of truncated Toeplitz operators*, Oper. Matrices 1 (2007), 491–526.
- [3] A. Baranov, I. Chalendar, E. Fricain, J. Mashreghi, and D. Timotin, *Bounded symbols and reproducing kernel thesis for truncated Toeplitz operators*, J. Funct. Anal. 259 (2010), 2673–2701; arXiv:0909.0131.
- [4] A. Baranov, R. Bessonov, and V. Kapustin, *Symbols of truncated Toeplitz operators*, J. Funct. Anal. 261 (2011), 3437–3456; arXiv:1009.5123.
- [5] P. Ahern and D. Clark, *Radial limits and invariant subspaces*, Amer. J. Math. 92 (1970), 332–342.
- [6] W. Cohn, *Radial limits and star invariant subspaces of bounded mean oscillation*, Amer. J. Math. 108 (1986), 719–749.